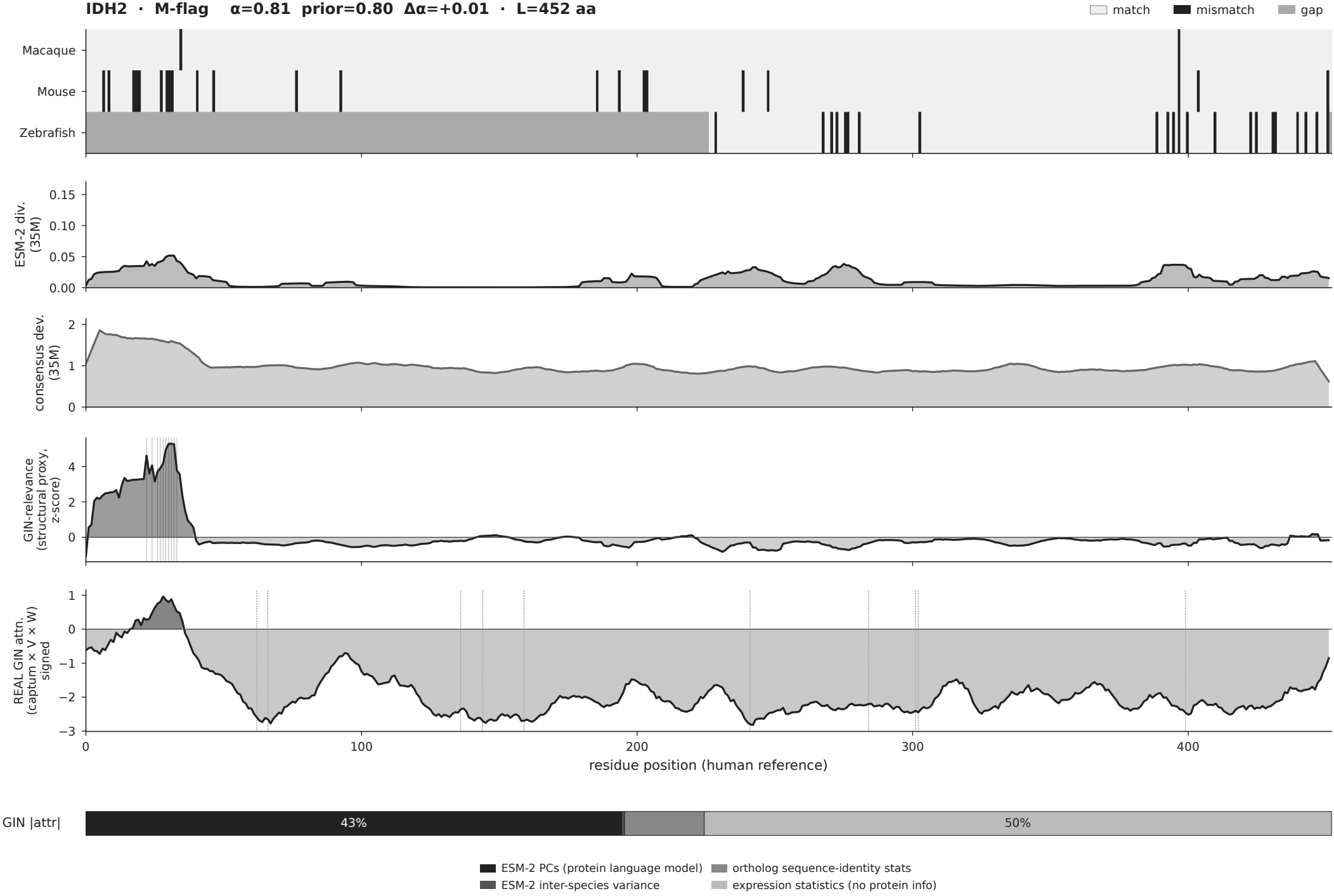
IDH2 · M-flag α=0.81 prior=0.80 Δα=+0.01 · L=452 aa
match
mismatch
gap
Macaque
Mouse
Zebrafish
ESM-2 div. (35M)
0.15
0.10
0.05
0.00
consensus dev. (35M)
2
1
0
GIN-relevance (structural proxy, z-score)
4
2
0
REAL GIN attn. (captum × V × W) signed
1
0
−1
−2
−3
0
100
200
300
400
residue position (human reference)
GIN |attr|
43%
50%
ESM-2 PCs (protein language model)
ortholog sequence-identity stats
ESM-2 inter-species variance
expression statistics (no protein info)